7/26 Experiment: 1 Basic Networking commands

1. arp -a
Address resolution protocol, shows the IP address of the computer along with IP and MAC address of the router.

```
PS C:\Users\ramch\Documents> arp -a

Interface: 192.168.29.59 --- 0x9
  Internet Address      Physical Address      Type
  192.168.29.1          8c-a3-99-d1-18-90     dynamic
  192.168.29.53         68-b9-c2-4e-c8-d8     dynamic
  192.168.29.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

2. hostname
Displays the name of the computer.

```
PS C:\Users\ramch\Documents> hostname
RAM
```

3. ipconfig /all
Displays the detailed network configuration information including IP address, MAC address, subnet mask, default gateway.

```
PS C:\Users\ramch\Documents> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : RAM
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : C2-BF-BE-87-96-64
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : C2-BF-BE-87-86-74
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

4. netstat:
Displays active network connections and ports.

```
PS C:\Users\ramch\Documents> netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    192.168.29.59:49271    ec2-3-120-237-193:https  CLOSE_WAIT
  TCP    192.168.29.59:51177    ec2-18-185-86-200:https  CLOSE_WAIT
  TCP    192.168.29.59:51178    151.101.157.55:https     ESTABLISHED
  TCP    192.168.29.59:51815    ec2-3-111-241-224:https  ESTABLISHED
  TCP    192.168.29.59:55103    ec2-35-173-153-86:https  ESTABLISHED
  TCP    192.168.29.59:55162    ec2-3-111-241-224:https  ESTABLISHED
  TCP    192.168.29.59:56751    ec2-16-144-18-157:https  CLOSE_WAIT
```

5. nslookup google.com
Finds the IP address of a domain name.

```
PS C:\Users\ramch\Documents> nslookup google.com
Server:  reliance.reliance
Address:  2405:201:e02e:2862::c0a8:1d01

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4013:813::71
          2404:6800:4013:813::65
          2404:6800:4013:813::64
          2404:6800:4013:813::8b
          142.251.221.174
```

6. pathping:
Combines the features of ping and tracert.
Shows the path taken by data packets to reach a destination and checks for packet loss and network delays at each step.

```
PS C:\Users\ramch\Documents> pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
   -g host-list      Loose source route along host-list.
   -h maximum_hops   Maximum number of hops to search for target.
   -i address        Use the specified source address.
   -n                Do not resolve addresses to hostnames.
   -p period         Wait period milliseconds between pings.
   -q num_queries    Number of queries per hop.
   -w timeout        Wait timeout milliseconds for each reply.
   -4                Force using IPv4.
   -6                Force using IPv6.
```

7. pathping google.com

```
PS C:\Users\ramch\Documents> pathping google.com

Tracing route to google.com [2404:6800:4013:813::8b]
over a maximum of 30 hops:
  0  RAM [2405:201:e02e:2862:bdc3:848a:f18:d6ef]
  1  2405:201:e02e:2862:8ea3:99ff:fed1:1890
  2     *        *        *
Computing statistics for 25 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  Address
  0                                           RAM [2405:201:e02e:2862:bdc3:848a:f18:d6ef]
                                 0/ 100 =  0%   |
  1   16ms     0/ 100 =  0%     0/ 100 =  0%   2405:201:e02e:2862:8ea3:99ff:fed1:1890

Trace complete.
```

3. ping google.com
checks computer can connect to Google server and measure the response time.

```
PS C:\Users\ramch\Documents> ping google.com

Pinging google.com [2404:6800:4007:80e::200e] with 32 bytes of data:
Reply from 2404:6800:4007:80e::200e: time=11ms
Reply from 2404:6800:4007:80e::200e: time=12ms
Reply from 2404:6800:4007:80e::200e: time=14ms
Reply from 2404:6800:4007:80e::200e: time=15ms

Ping statistics for 2404:6800:4007:80e::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 15ms, Average = 13ms
```

- tracert google.com
shows the path taken by data packets to reach a destination.

```
PS C:\Users\ramch\Documents> tracert google.com

Tracing route to google.com [2404:6800:4007:80e::200e]
over a maximum of 30 hops:

  1     7 ms     5 ms     4 ms  2405:201:e02e:2862:8ea3:99ff:fed1:1890
  2     *        *        *     Request timed out.
  3    29 ms    10 ms     9 ms  2405:203:400:100:172:31:0:144
  4     *        *        *     Request timed out.
  5    32 ms    10 ms    11 ms  2405:200:80c:3168:61::5
  6     *        *        *     Request timed out.
  7    25 ms    11 ms    12 ms  2405:200:801:900::1294
```

2. ipconfig
displays the IP configuration of a computer

```
PS C:\Users\ramch\Documents> ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

- ipconfig /renew:
Request a new IP address from the DHCP server for the your network adapter, used to refresh from the current IP configuration

```
PS C:\Users\ramch\Documents> ipconfig /renew

Windows IP Configuration

No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

12. ipconfig/displaydns - Displays the contents of the local DNS resolver cache, showing recently resolved domain names and their IP addresses.

```
PS C:\Users\ramch\Documents> ipconfig /displaydns

Windows IP Configuration

    telemetry-incoming.r53-2.services.mozilla.com
    ----------------------------------------
    Record Name . . . . . : telemetry-incoming.r53-2.services.mozilla.com
    Record Type . . . . . : 1
    Time To Live  . . . . : 1025
    Data Length . . . . . : 4
    Section . . . . . : Answer
    A (Host) Record . . . : 34.120.208.123

    200.86.185.18.in-addr.arpa
    ----------------------------------------
```

13. netsh wlan show interfaces - Display info about the wireless interface, including the connected SSID, signal strength, radio types and connection states.

```
PS C:\Users\ramch\Documents> netsh wlan show interface

There is 1 interface on the system:

    Name                   : Wi-Fi
    Description            : MediaTek Wi-Fi 6E MT7922 (RZ616) 160MHz PCIe Adapter
    GUID                   : 67a10457-be9e-4090-8f86-ba97ee3f41a1
    Physical address       : c0:bf:be:87:b6:44
    Interface type         : Primary
    State                  : connected
    SSID                   : GANESHGOPIKA_5G
```

14. netsh wlan show profiles : Display a list of all wifi network profiles saved on the computer.

```
PS C:\Users\ramch\Documents> netsh wlan show profiles

Profiles on interface Wi-Fi:

Group policy profiles (read only)
---------------------------------
    <None>

User profiles
-------------
    All User Profile     : GANESHGOPIKA_5G
    All User Profile     : iQOO
    All User Profile     : A_New_104
    All User Profile     : SANGILI POOTHATHAN
    All User Profile     : SSRT
```

5. route print

Display the computers routing table, showing how network traffic is routed to different destinations.

```
PS C:\Users\ramch\Documents> route print
===========================================================================
Interface List
 11...c2 bf be 87 96 64 ......Microsoft Wi-Fi Direct Virtual Adapter
  8...c2 bf be 87 86 74 ......Microsoft Wi-Fi Direct Virtual Adapter #2
  9...c0 bf be 87 b6 44 ......MediaTek Wi-Fi 6E MT7922 (RZ616) 160MHz PCIe Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0    192.168.29.1    192.168.29.59     35
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
```

6. getmac

Displays the MAC address of computer.

```
PS C:\Users\ramch\Documents> getmac

Physical Address    Transport Name
=================== ==========================================================
C0-BF-BE-87-B6-44   \Device\Tcpip_{67A10457-BE9E-4090-8F86-BA97EE3F41A1}
```

17. ipconfig /release —

Release the current IP address

```
PS C:\Users\ramch\Documents> ipconfig /release

Windows IP Configuration

No operation can be performed on Local Area Connection* 1 while it has its media disconnected.
No operation can be performed on Local Area Connection* 2 while it has its media disconnected.

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

18. ~~netstat~~ -an — Display all active network connections.

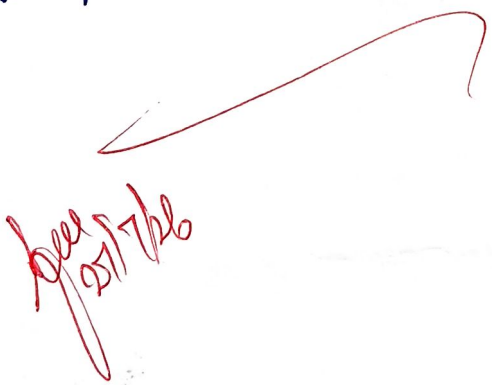
```
PS C:\Users\ramch\Documents> netstat -an

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING
  TCP    0.0.0.0:49669          0.0.0.0:0              LISTENING
  TCP    169.254.222.184:139    0.0.0.0:0              LISTENING
  TCP    192.168.29.59:58807    20.190.145.143:443     TIME_WAIT
```

Linux Commands:

1. ip addr - Display IP address and network interface details.
2. ip link - Show the status of network interfaces.
3. dig - Provide detailed DNS information.
4. ss - Display active network connections & listening ports
5. arp - Display ARP table info.



Apeel 23/7/26